Sudoku Solver

5	3		6	7	8	9	.	2
6	7	2	1	9	5	3	4	8
	9	8	3	4	2	5	6	7
8	5	9	7	6	1	4	2	3
4	2	6	8	5	3	7	9	1
7	1	3	9		4	8	5	6
9	6		5	3	7	2	8	4
2	8	7	4	1	9	6	3	5
3	4	5	2	8	6	1	7	9

Conditions

1.) 1 – 9 once in every row
2.) 1 – 9 once in every column
3.) 1 – 9 once in every 3×3 grid

True

False

T

5	3		6	7	8	9		2
6	7	2	1	9	5	3	4	8
	9	8	3	4	2	5	6	7
8	5	9	7	6	1	4	2	3
4	2	6	8	5	3	7	9	1
7	1	3	9		4	8	5	6
9	6		5	3	7	2	8	4
2	8	7	4	1	9	6	3	5
3	4	5	2	8	6	1	7	9

5	3	1	6	7	8	9		2
6	7	2	1	9	5	3	4	8
	9	8	3	4	2	5	6	7
8	5	9	7	6	1	4	2	3
4	2	6	8	5	3	7	9	1
7	1	3	9		4	8	5	6
9	6		5	3	7	2	8	4
2	8	7	4	1	9	6	3	5
3	4	5	2	8	6	1	7	9

5	3	1	6	7	8	9	1	2
6	7	2	1	9	5	3	4	8
	9	8	3	4	2	5	6	7
8	5	9	7	6	1	4	2	3
4	2	6	8	5	3	7	9	1
7	1	3	9		4	8	5	6
9	6		5	3	7	2	8	4
2	8	7	4	1	9	6	3	5
3	4	5	2	8	6	1	7	9

T

5	3	1	6	7	8	9	1	2
6	7	2	1	9	5	3	4	8
1	9	8	3	4	2	5	6	7
8	5	9	7	6	1	4	2	3
4	2	6	8	5	3	7	9	1
7	1	3	9		4	8	5	6
9	6		5	3	7	2	8	4
2	8	7	4	1	9	6	3	5
3	4	5	2	8	6	1	7	9

T

5	3	1	6	7	8	9	1	2
6	7	2	1	9	5	3	4	8
1	9	8	3	4	2	5	6	7
8	5	9	7	6	1	4	2	3
4	2	6	8	5	3	7	9	1
7	1	3	9	2	4	8	5	6
9	6		5	3	7	2	8	4
2	8	7	4	1	9	6	3	5
3	4	5	2	8	6	1	7	9

T

5	3	1	6	7	8	9	1	2
6	7	2	1	9	5	3	4	8
1	9	8	3	4	2	5	6	7
8	5	9	7	6	1	4	2	3
4	2	6	8	5	3	7	9	1
7	1	3	9	2	4	8	5	6
9	6	1	5	3	7	2	8	4
2	8	7	4	1	9	6	3	5
3	4	5	2	8	6	1	7	9

T

5	3	1	6	7	8	9	1	2
6	7	2	1	9	5	3	4	8
1	9	8	3	4	2	5	6	7
8	5	9	7	6	1	4	2	3
4	2	6	8	5	3	7	9	1
7	1	3	9	2	4	8	5	6
9	6	1	5	3	7	2	8	4
2	8	7	4	1	9	6	3	5
3	4	5	2	8	6	1	7	9

Pseudocode

```
for (i → rows)
{for (j → cols)
{
    if (board[i][j] == '.')
    {
        for (c → 1 to 9)
        {
            isvalid(board, i, j, c)
            {
                board[i][j] = c
                if (solve(board))
                    return true
                else
                board[i][j] = '.'
            }
        }
    } return false;
}
return true;
```

```
isValid(c)
{
    // check if c is already present in the row
        return false;
    // check if c is already present in the col
        return false;
    // check if c is already present in 3x3 grid
        return false;

}
```